

HOMework 8 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

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Due at 10:00 EST in class

Q1 Solve the following using the superposition method:

(a) $y'' + y = 5 \cos(2x)$

(b) $y'' - y = xe^x$

(c) $y'' + 4y' + 4y = e^{-2x} + x$

Q2 For each of the following, determine the homogeneous solution and the **form** (do not solve) of a particular non-homogeneous solution:

(i) $y'' + 4y = 2x^2$

(ii) $y'' + 4y = 10 + \sin(x)$

(iii) $y'' + 4y = \sin(2x)$

(iv) $y'' + 4y = e^x \sin(2x)$

Q3 Find the general solution solution of $y'' + 4y = x \cos x$ using the variation of parameters approach.